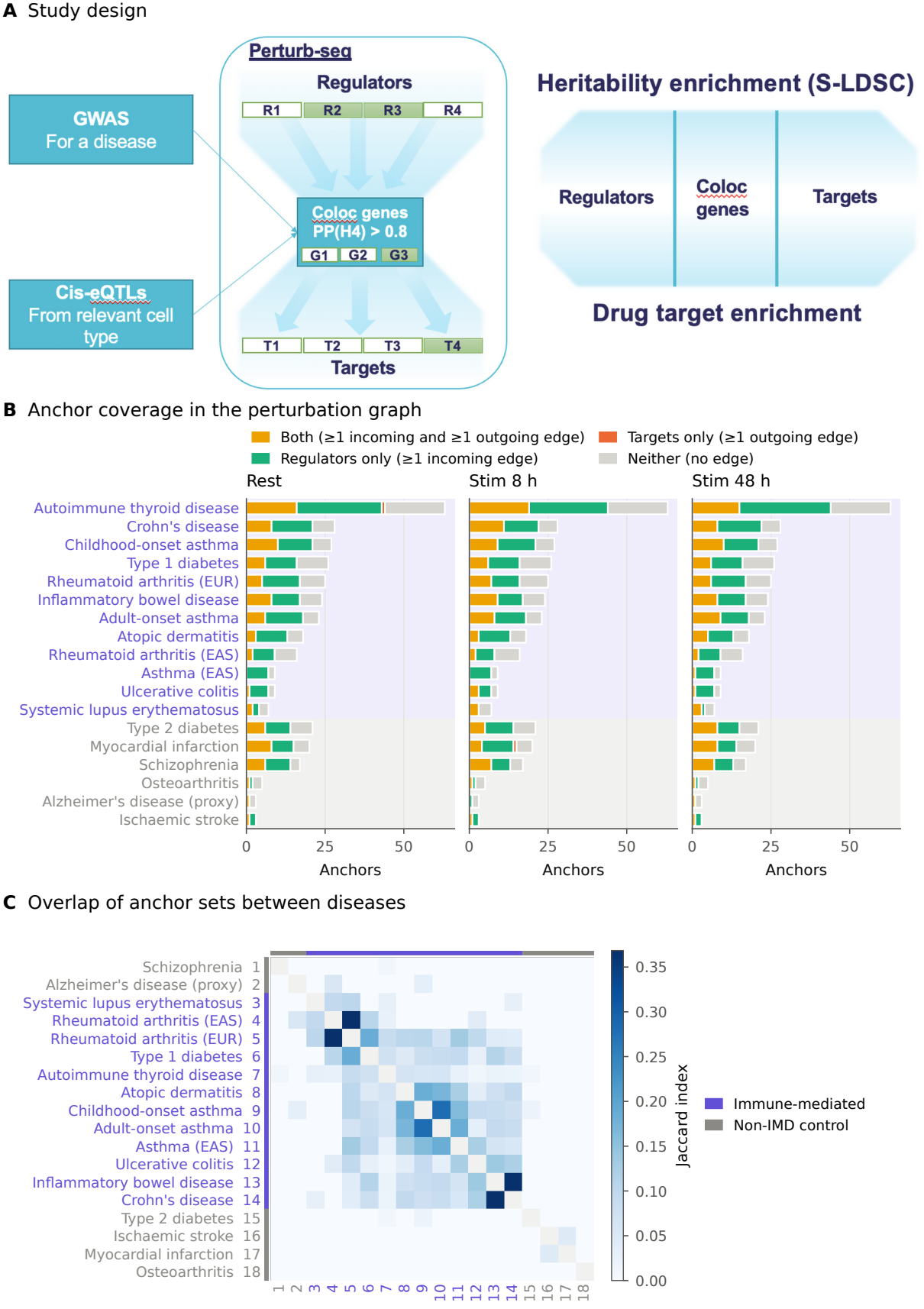

A Study design
GWAS
For a disease
Cis-eQTLs
From relevant cell type
Perturb-seq
Regulators
R1
R2
R3
R4
Coloc genes
PP(H4) > 0.8
G1
G2
G3
T1
T2
T3
T4
Targets
Heritability enrichment (S-LDSC)
Regulators
Coloc genes
Targets
Drug target enrichment
B Anchor coverage in the perturbation graph
Both (≥1 incoming and ≥1 outgoing edge)
Targets only (≥1 outgoing edge)
Regulators only (≥1 incoming edge)
Neither (no edge)
Rest
Stim 8 h
Stim 48 h
Autoimmune thyroid disease
Crohn's disease
Childhood-onset asthma
Type 1 diabetes
Rheumatoid arthritis (EUR)
Inflammatory bowel disease
Adult-onset asthma
Atopic dermatitis
Rheumatoid arthritis (EAS)
Asthma (EAS)
Ulcerative colitis
Systemic lupus erythematosus
Type 2 diabetes
Myocardial infarction
Schizophrenia
Osteoarthritis
Alzheimer's disease (proxy)
Ischaemic stroke
0
25
50
Anchors
C Overlap of anchor sets between diseases
Schizophrenia 1
Alzheimer's disease (proxy) 2
Systemic lupus erythematosus 3
Rheumatoid arthritis (EAS) 4
Rheumatoid arthritis (EUR) 5
Type 1 diabetes 6
Autoimmune thyroid disease 7
Atopic dermatitis 8
Childhood-onset asthma 9
Adult-onset asthma 10
Asthma (EAS) 11
Ulcerative colitis 12
Inflammatory bowel disease 13
Crohn's disease 14
Type 2 diabetes 15
Ischaemic stroke 16
Myocardial infarction 17
Osteoarthritis 18
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Jaccard index
Immune-mediated
Non-IMD control